

Plant Whisperer: AI-Driven Agricultural Intelligence for Saffron Cultivation

Precision Farming for High-Value Crops



\$200B+

Global Agricultural Losses.



Extreme Environmental Sensitivity: Timing, Moisture, Climate.



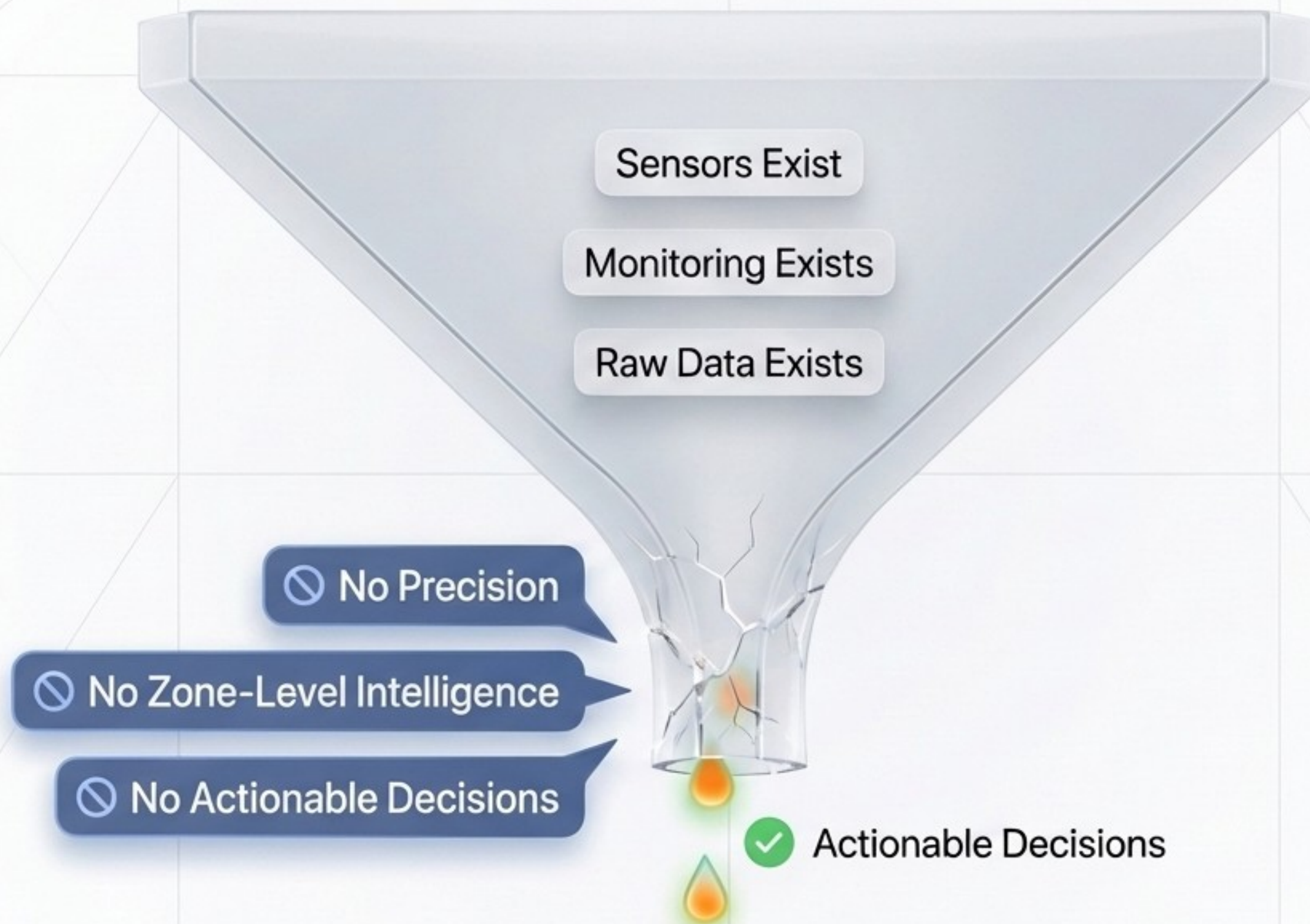
Single-Issue Devastation.

₹20,000–₹1,00,000

financial loss per acre from a single miscalculation.

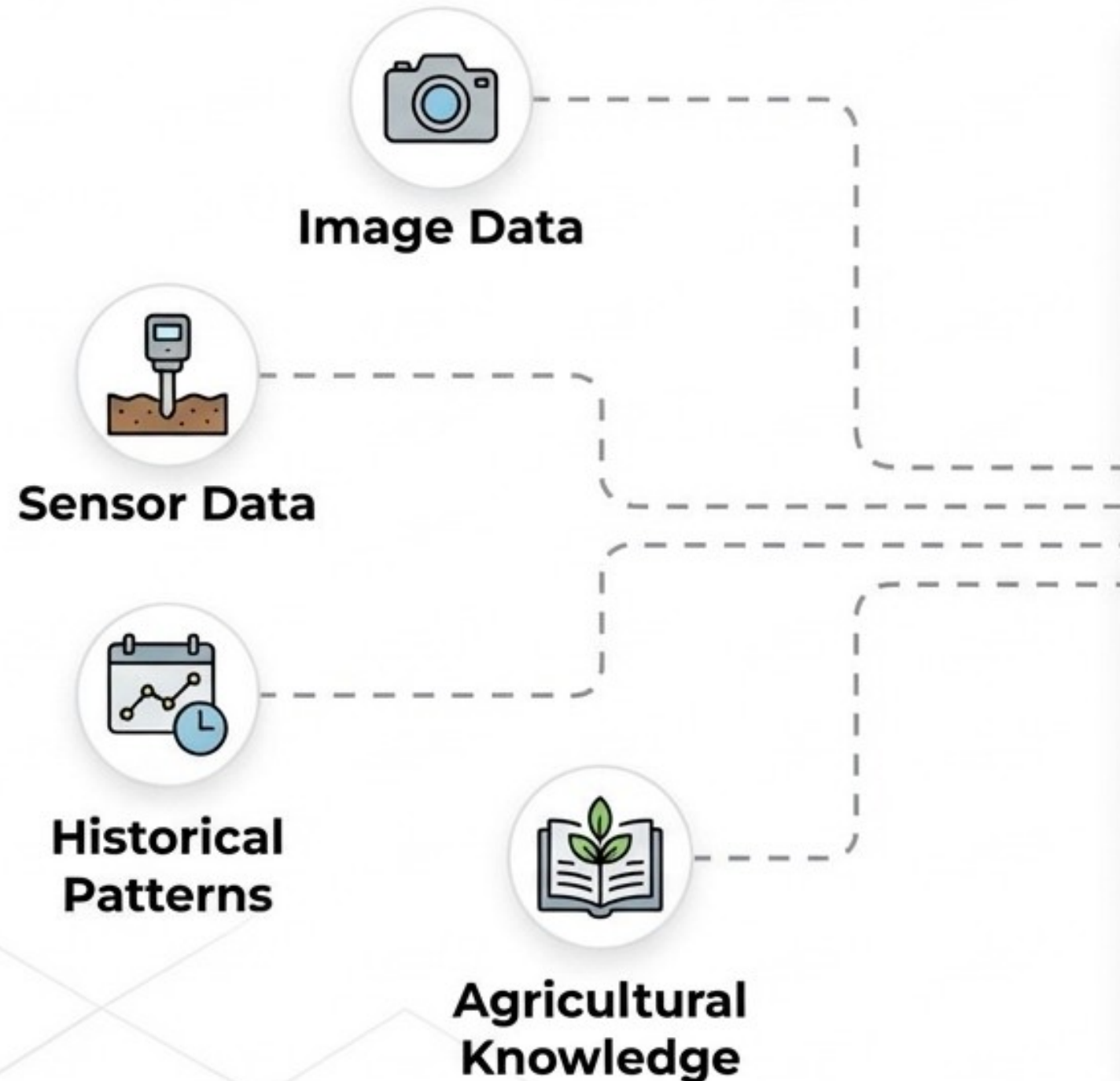
Small inefficiencies trigger massive financial loss.

Existing Infrastructure



Agriculture has intelligence, but lacks precision.

The Precision Agriculture Gap



Problem 1: Bold

No system integrates these disparate data streams.



Problem 2: Bold

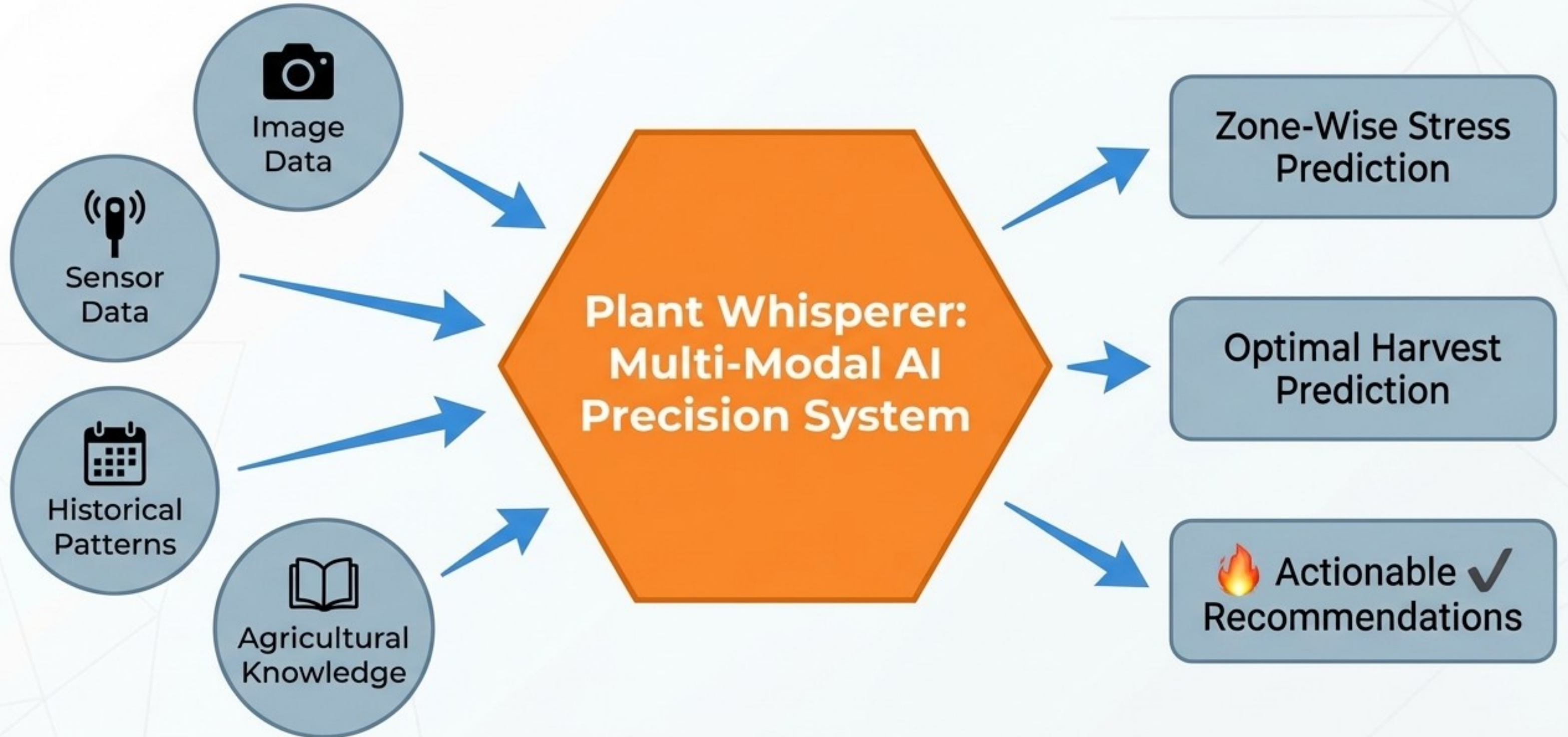
No system provides context-aware, zone-wise recommendations.



Problem 3: Bold

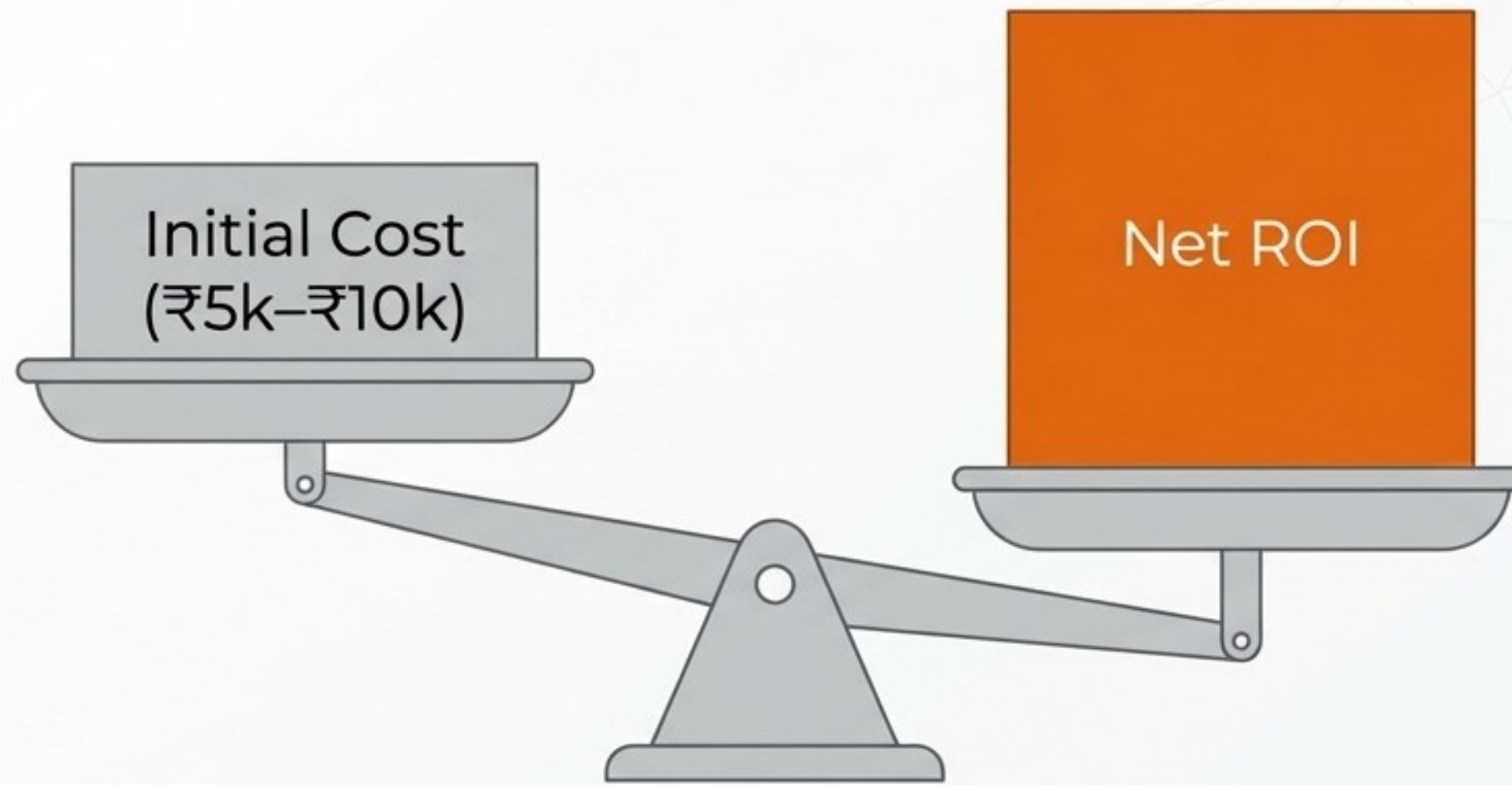
Current tools ignore profit-driven decision-making.

SF Pro Display Ultra-Bold



We are not selling technology — we are selling increased profit.

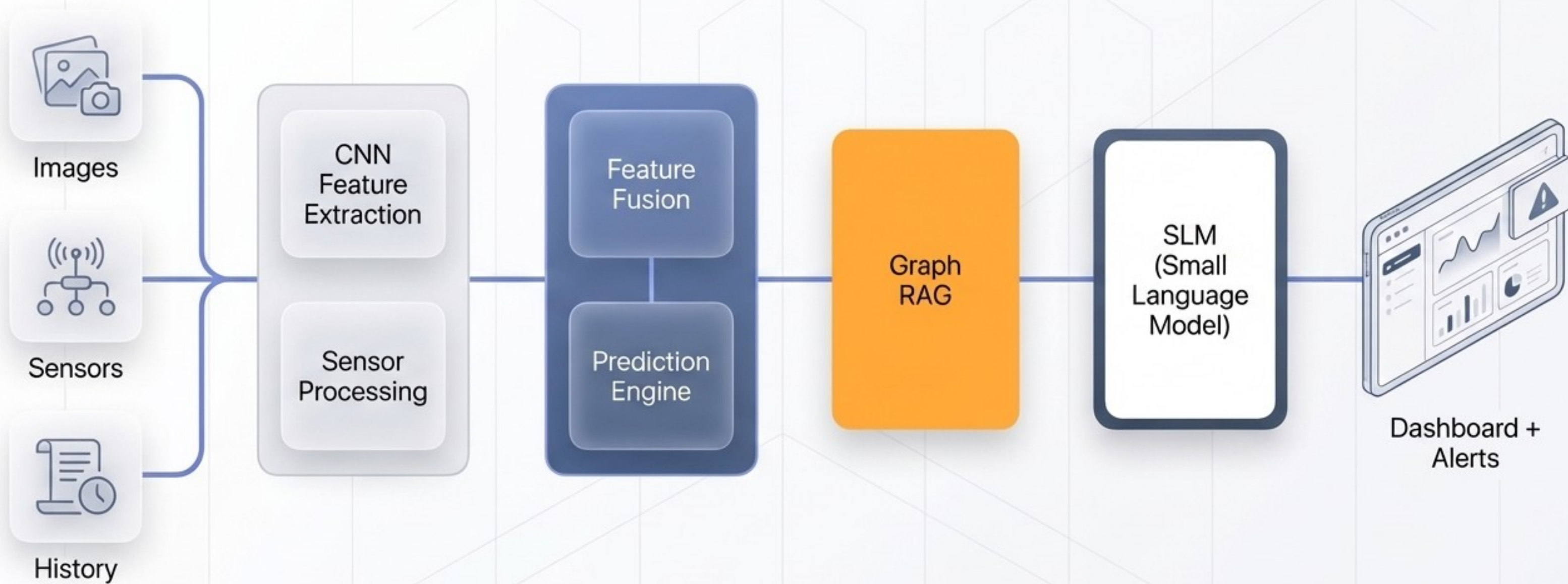




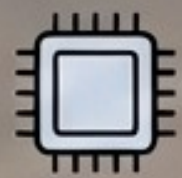
Gains exceed operational costs within the first harvest cycle.
The system is fundamentally ROI-positive.

The system does not add cost — it converts environmental uncertainty into predictable profit.

End-to-End System Architecture



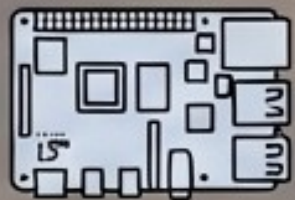
Edge Infrastructure



ESP32



Sensors

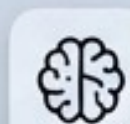


Raspberry Pi

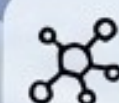


Image
Processing

Cloud Architecture



ML Models

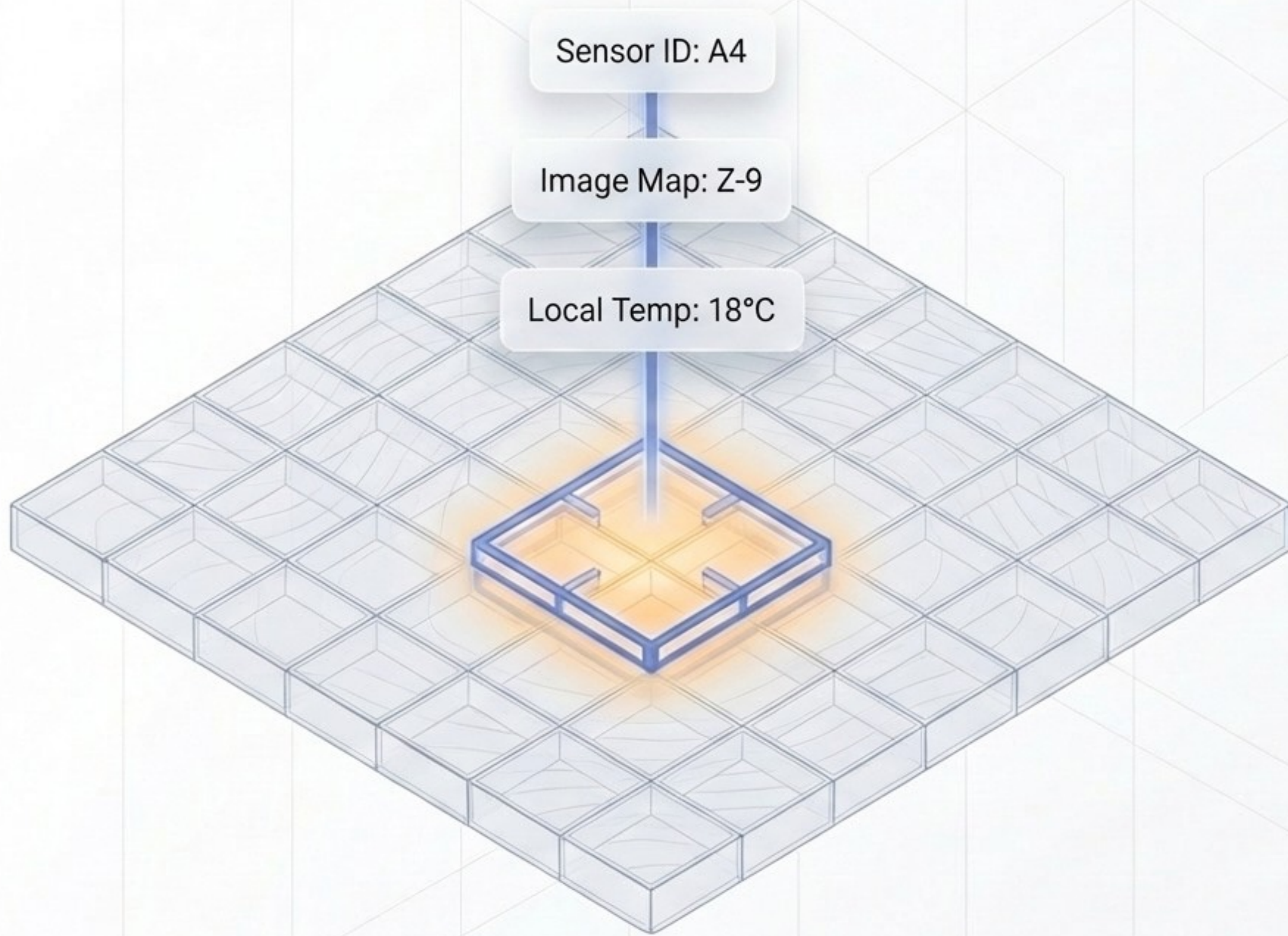


Knowledge
Graph

SLM

SLM

A hybrid architecture minimizing edge bandwidth while maximizing cloud intelligence.



Grid-Based Precision System

Each micro-grid acts as an independent analytical unit. The system generates

The system generates micro-zone intelligence rather than relying on flawed whole-farm averages.

Stress Model

$$S_{\text{final}} = (w_1 \times \text{📷}) + (w_2 \times \text{☁️☀️}) + (w_3 \times \text{🕒})$$

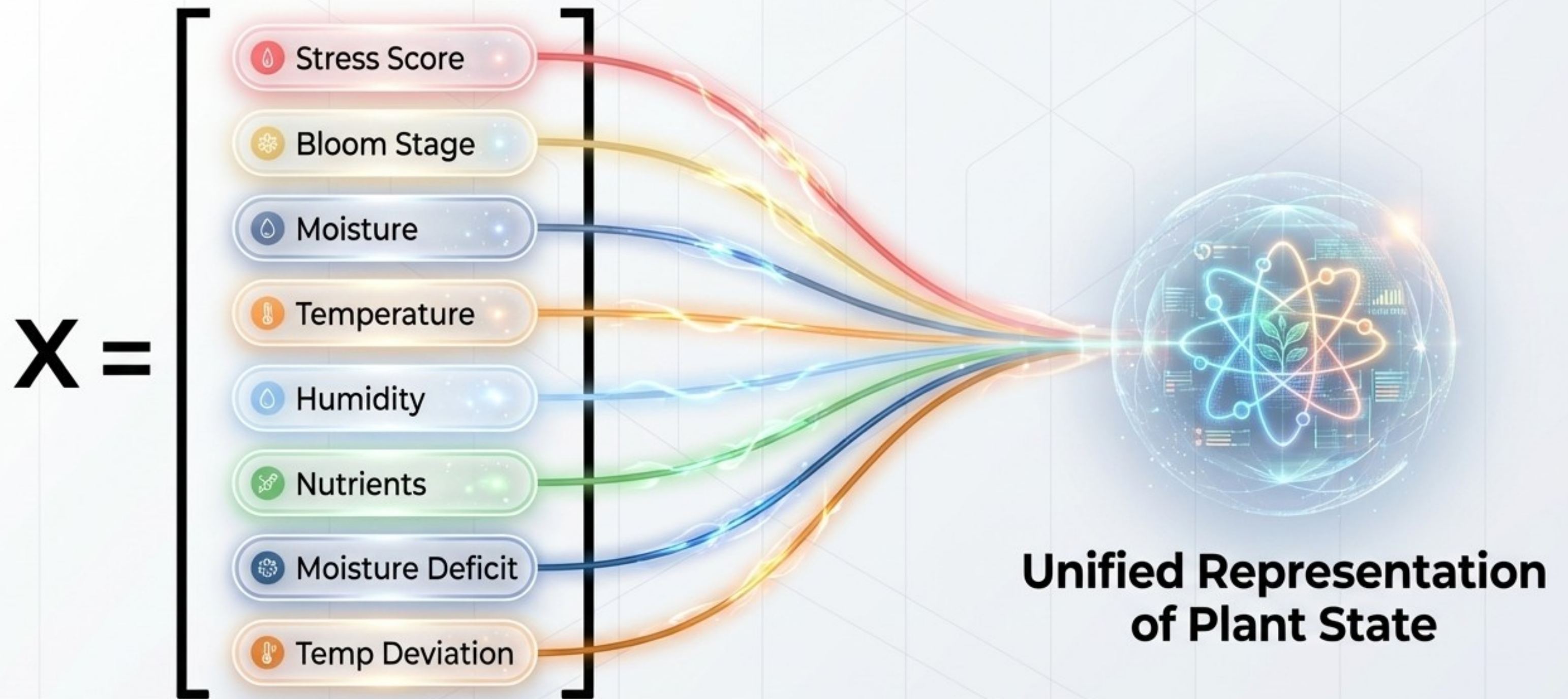
Interpretable weighting of visual, environmental, and historical stress.

Harvest Model

$$T_{\text{adjusted}} = T_{\text{base}} - (\alpha \times \text{🍃}) - (\beta \times \text{🌡️}) + (\gamma \times \text{⚠️})$$

Dynamic harvest timing adjustment based on real-time physiological data.

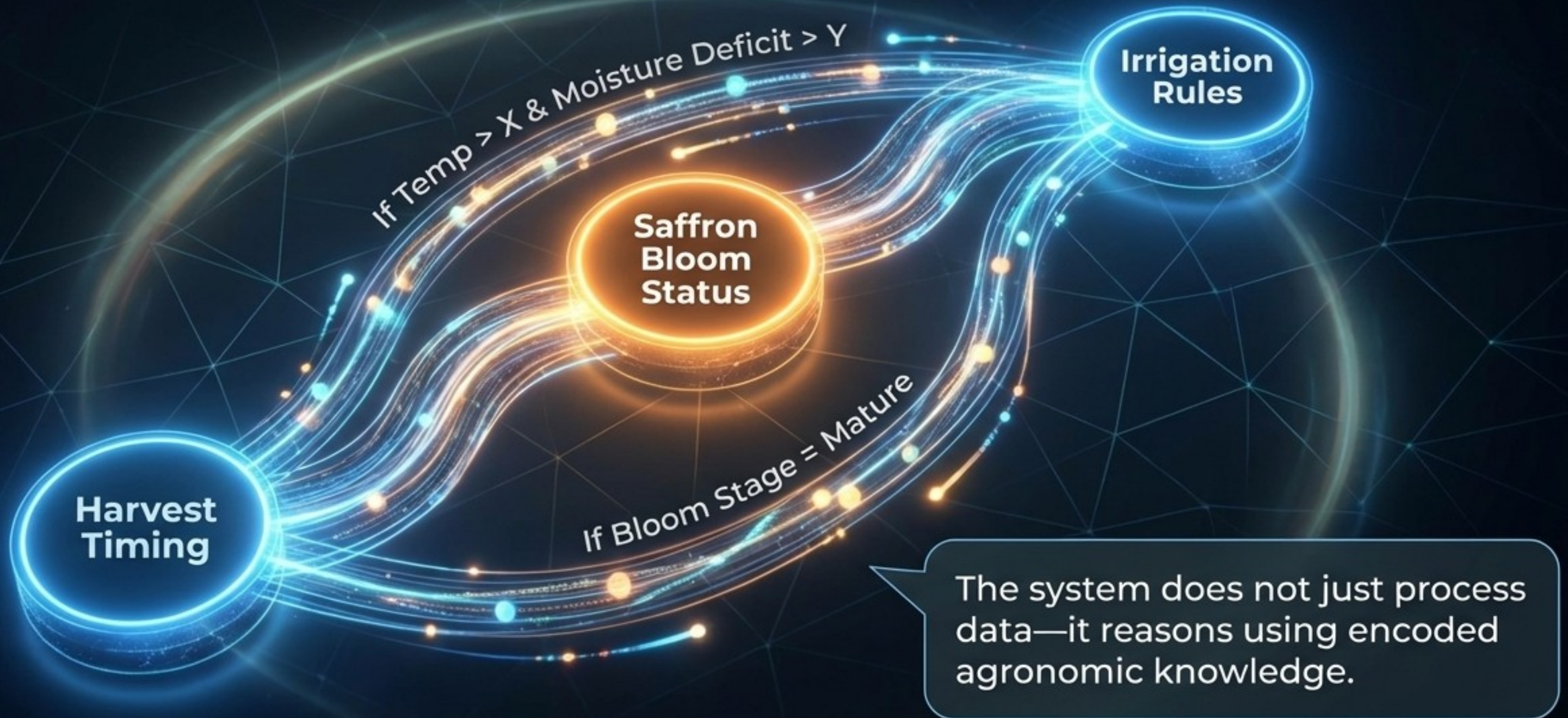
A hybrid approach combining the predictability of interpretable formulas with the adaptability of learned models.



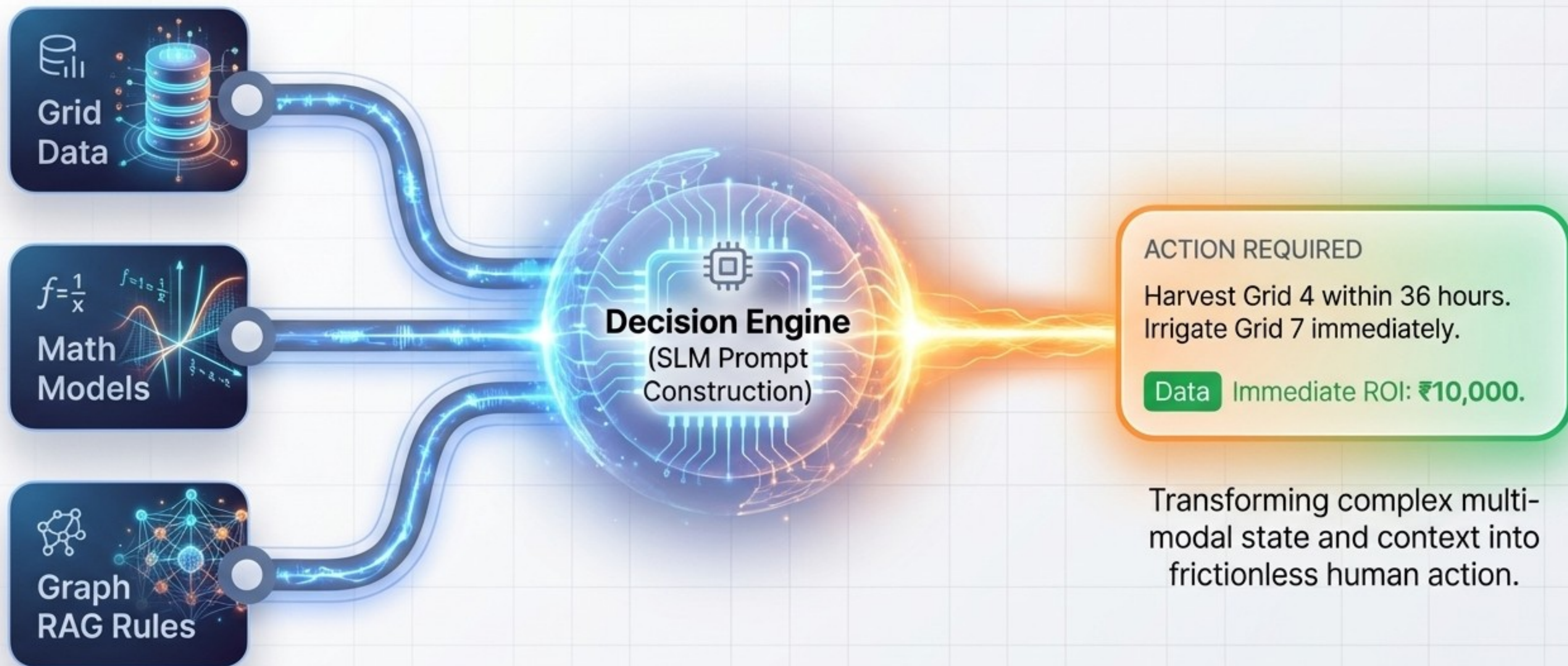
Feature Fusion transforms isolated metrics into a singular, holistic physiological profile.

Core Intelligence: Graph RAG System

The knowledge graph is built directly from peer-reviewed agricultural documents.



Data Culminates in Direct Economic Action



The Unfair Advantage in AgTech.

Feature	Traditional Systems	Plant Whisperer
Resolution Level	 Whole Farm	☒ Micro-Grid Precision 
Analytical Engine	 Static Thresholds	☒ Graph RAG Reasoning 
Decision Output	 Spreadsheets	☒ SLM Action Prompts 
Primary Metric	 Data Logging	☒ Net ROI & Profit 

Cost-efficient SLM architecture | Shared drone monitoring model (Satellite + Drone hybrid)